TokTickIT - Sprint 4 Test Plan and Traceability

Written alongside the Sprint 4 contract, before the implementation PRs (Test DD/TDD). Every Acceptance Criterion in `specification.md` maps to at least one row below. The **Final** column is `Planned` until the corresponding test is implemented and run against `main`, at which point it is updated to `Pass` / `Fail` for submission (Part 3 requires the actual passing output from `main`).

0. Coverage Summary

Category	Count	Primary file(s)
Unit	18	`server/tests/lab-04/transition-matrix.api.test.ts`
API / Integration	20	`server/tests/lab-04/*.api.test.ts`
UI Component	9	`client/tests/lab-04/*.test.tsx`
UI Style	2	`client/tests/lab-04/ActionsTaken.test.tsx`
Responsive	2	`client/tests/lab-04/ActionsTaken.test.tsx`, `client/tests/lab-04/StaffDashboard.test.tsx`
Security / Authorization	4	`server/tests/lab-04/actions-taken.api.test.ts`, `server/tests/lab-04/ticket-workflow.api.test.ts`, `server/tests/lab-04/admin-dashboard.api.test.ts`
Migration / Regression	1	`server/tests/lab-04/actions-taken.api.test.ts`
End-to-End	5	`e2e/lab-04/*.spec.ts`

Authorization and migration counts overlap the API / Integration count. The UI component count excludes the two style and two responsive tests, which are listed separately. Action field validation is covered through API integration tests; the repository does not currently have a separate Action Taken validator unit-test file.

1. Unit

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
UNIT-04	Unit	BR-09	Status-transition matrix lookup, role-specific next statuses, self-transitions, and unknown status inputs	Every documented edge is allowed; undocumented/invalid edges are rejected; only permitted roles receive the corresponding next status	`server/tests/lab-04/transition-matrix.api.test.ts`	Pass
UNIT-05	Unit	BR-07	Resolved-gate check with no actions, non-Resolved results, staff/admin results, requester results, and case variants	The gate opens only for an exact `Resolved` result recorded by IT Staff or Administrator	`server/tests/lab-04/transition-matrix.api.test.ts`	Pass

2. API / Integration

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Fina
API-01	API	FR-05, BR-03	`performedById` sent by client on create	Ignored; server sets it from session regardless of body content	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-02	API	BR-16	Create Action Taken on a `Closed` Ticket	`409 Conflict`	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-03	API	AC-01	Create a valid Action Taken	Created under the correct Ticket and actor	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-04	API	AC-03, BR-04	Create with `followUpRequired=true` and empty note	Validation error identifies `followUpNote`; no record is created	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-05	API	AC-07, BR-11	Edit Action Taken with stale `updatedAt`	`409 Conflict`, current record returned, no data overwritten	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-06	API	AC-05, AC-06, BR-07	Transition to `Resolved` with/without a staff-recorded `Resolved` Action Taken	Rejected without one; succeeds with one	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pas
API-07	API	BR-09, BR-11	Invalid transition, unknown status, and stale Ticket `updatedAt`	Invalid requests are rejected; stale update returns a conflict without overwriting	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pas
API-08	API	AC-02, BR-14	`GET /api/requester/dashboard` with a spoofed/foreign requester id in the query	Own session's data only; foreign id ignored; recent Tickets belong to the session user	`server/tests/lab-04/requester-dashboard.api.test.ts`	Pas
API-09	API	AC-08, BR-12, BR-13	`GET /api/staff/dashboard` counts against a known seeded dataset	Unassigned/My Assigned/status/priority counts match an independently computed expectation	`server/tests/lab-04/staff-dashboard.api.test.ts`	Pas
API-10	API	API spec §1	Omit `actionAt` when creating an Action Taken	Database current-time default is used	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-11	API	FR-02	Requester lists Actions Taken for their own Ticket	`200`; own Ticket's actions are available read-only	`server/tests/lab-04/actions-taken.api.test.ts`	Pas
API-12	API	FR-11, API spec §5	IT Staff requests the Administrator Dashboard	`403`; no staff dashboard or user-count data is returned	`server/tests/lab-04/admin-dashboard.api.test.ts`	Pas

3. UI Component

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
UI-01	UI	FR-04	`ActionsTaken` renders for a Requester session	List is read-only; no Add/Edit controls present in the DOM	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass
UI-02	UI	AC-11	`ActionsTaken` with zero records	Renders the defined empty state, no error	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass
UI-03	UI	AC-13	Rapid double-click on Save in the create form	Submit button disables after first click; only one request fires	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass
UI-04	UI	FR-06, AC-05	Ticket without a Resolved-gate action	`Resolved` option is disabled and the reason is explained; no transition request is sent	`client/tests/lab-04/TicketWorkflow.test.tsx`	Pass
UI-05	UI	AC-09	Requester Dashboard with zero Tickets	All four cards show `0`, no error state	`client/tests/lab-04/RequesterDashboard.test.tsx`	Pass
UI-06	UI	FR-12, AC-10	Clicking the Unassigned staff metric	Navigates to Ticket Queue with the matching owner filter applied	`client/tests/lab-04/StaffDashboard.test.tsx`	Pass
UI-07	UI	FR-02, FR-04	Actions Taken endpoint selected for IT Staff, Administrator, and Requester	Each role loads the Ticket's actions from its role-appropriate API endpoint	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass

4. UI Style / Visual Consistency

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
STYLE-01	UI style	ui-spec.md §7	Actions Taken styles and class conventions	Existing Zen Green tokens and established style conventions are used	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass
STYLE-02	UI style	ui-spec.md §4, §7	Result badges and follow-up tags	Color is paired with visible text/non-color cues	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass

5. Responsive

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
RESP-01	Responsive	ui-spec.md §2, §3, §6	Dashboard card grids at desktop/tablet/mobile breakpoints	Six- and four-card grids reflow to the documented tablet and mobile layouts	`client/tests/lab-04/StaffDashboard.test.tsx`	Pass
RESP-02	Responsive	ui-spec.md §4, §6	Actions Taken table at mobile width	Table collapses to stacked cards with per-field labels	`client/tests/lab-04/ActionsTaken.test.tsx`	Pass

6. Authorization

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
AUTH-01	Auth	AC-04	Requester calls `POST /api/tickets/:id/actions` directly	`403 Forbidden`	`server/tests/lab-04/actions-taken.api.test.ts`	Pass
AUTH-02	Auth	BR-10, AC-12	IT Staff attempts to exit a `Cancelled` / `Closed` Ticket	`403 Forbidden`; Administrator may reopen	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
AUTH-03	Auth	§5 api-spec.md	Authorization sweep for status update, resolution, and Action Taken creation across anonymous, Requester, IT Staff, and Administrator callers	Each tested request matches its documented role/access rule	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
AUTH-04	Auth	FR-11, API spec §5	IT Staff attempts to access the Administrator Dashboard	`403 Forbidden` and no dashboard payload is leaked	`server/tests/lab-04/admin-dashboard.api.test.ts`	Pass
AUTH-05	Auth	FR-02, API spec §1	Requester requests Actions Taken for another Requester's Ticket	`403 Forbidden`; no foreign Ticket action data is returned	`server/tests/lab-04/actions-taken.api.test.ts`	Pass

7. Workflow

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
WF-01	Workflow	BR-09	Every edge in the transition matrix	Documented edges succeed for allowed roles and are rejected for others	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
WF-02	Workflow	BR-08	Requester "appears resolved" flag alone	Flag is advisory; Ticket status does not change	`server/tests/lab-04/ticket-workflow.api.test.ts`	Pass
WF-03	Workflow	FR-06, AC-05, AC-06	Full lifecycle New → Open → In Progress → Resolved → Closed, adding a Resolved action before the gate	Valid transitions succeed in order; resolving is unavailable until the action is recorded	`e2e/lab-04/ticket-resolution.spec.ts`	Pass
WF-04	Workflow	BR-10, BR-16, AC-12	Administrator reopens a `Closed` Ticket and resumes normal work	Reopen succeeds for Admin; after returning to an active status, a new Action Taken can be added	`e2e/lab-04/ticket-resolution.spec.ts`	Pass

8. Migration

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
MIG-01	Migration	—	Additive migration run against an existing database	Existing records are unchanged; existing Tickets have zero Action Taken rows at migration time	`server/tests/lab-04/actions-taken.api.test.ts`	Pass

9. Performance smoke

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
PERF-01	Perf smoke	BR-12	`GET /api/staff/dashboard` against ~500 seeded Tickets	Responds in under 500 ms locally with correct counts	`server/tests/lab-04/staff-dashboard.api.test.ts`	Pass
PERF-02	Perf smoke	BR-12	`GET /api/requester/dashboard` against a Requester with ~50 Tickets	Responds in under 500 ms locally; `recentTickets` remains capped at 5	`server/tests/lab-04/requester-dashboard.api.test.ts`	Pass

10. End-to-End

Test ID	Type	Req/AC	What It Tests	Expected Result	Automated Test File	Final
E2E-01	E2E	AC-01	Full Actions Taken flow: IT Staff opens a Ticket, adds an action, sees it in the list, and edits it	Action appears immediately and the edit persists	`e2e/lab-04/actions-taken-flow.spec.ts`	Pass
E2E-02	E2E	AC-03	Create an Action Taken with Follow-Up Required checked and Follow-up Note left blank	Inline validation error shown; record is not created until the note is filled in	`e2e/lab-04/actions-taken-flow.spec.ts`	Pass
E2E-03	E2E	AC-05, AC-06, AC-10	Resolution gate end to end, then dashboard drill-down from both roles	Resolve is rejected until a Resolved action exists; requester/staff metrics and drill-downs reflect the status change	`e2e/lab-04/dashboards.spec.ts`	Pass

Acceptance Criteria → Test Traceability

AC	Covered by
AC-01	API-03, E2E-01
AC-02	API-08
AC-03	API-04, E2E-02
AC-04	AUTH-01

AC	Covered by
AC-05	API-06, E2E-03
AC-06	API-06, E2E-03
AC-07	API-05
AC-08	API-09
AC-09	UI-05
AC-10	UI-06, E2E-03
AC-11	UI-02
AC-12	AUTH-02, WF-04
AC-13	UI-03, E2E-01

```
Running 5 tests using 1 worker

  ✓  1 …ions Taken › E2E-01: IT Staff can add, view, and edit an Action Taken (10.8s)
  ✓  2 …4 Actions Taken › E2E-02: Follow-Up Required requires a follow-up note (8.3s)
  ✓  3 …olution gate unlocks and requester/staff dashboard drill-downs update (15.1s)
  ✓  4 …F-03: completes New → Open → In Progress → Resolved → Closed lifecycle (9.5s)
  ✓  5 …› WF-04: Administrator reopens a Closed Ticket and normal flow resumes (4.7s)
  5 passed (1.1m)


npm notice run toktickit-server@1.0.0 test
npm notice run vitest run tests/lab-04

 RUN  v5.0.1 D:/KMUTT/Year 3/Software Engineer/server

 ✓ tests/lab-04/ticket-workflow.api.test.ts (6 tests) 1643ms
   ✓ Lab 4 Ticket Workflow API (6)
     ✓ WF-01: every matrix edge succeeds for allowed roles and is rejected for others 524ms
 ✓ tests/lab-04/actions-taken.api.test.ts (10 tests) 748ms
 ✓ tests/lab-04/requester-dashboard.api.test.ts (2 tests) 616ms
 ✓ tests/lab-04/staff-dashboard.api.test.ts (2 tests) 499ms
 ✓ tests/lab-04/transition-matrix.api.test.ts (18 tests) 8ms
 ✓ tests/lab-04/admin-dashboard.api.test.ts (1 test) 292ms

 Test Files  6 passed (6)
      Tests  39 passed (39)
   Start at  09:17:37
   Duration  5.54s (tests 77%, import 18%, transform 4%)

    Isolate  6 workers spawned · ~95ms startup each (spawn + environment, per file)
             at least ~473ms faster with isolate: false — reuses workers across files instead of one per file
```

```
npm notice run vitest run tests/lab-04

 RUN  v2.1.9 D:/KMUTT/Year 3/Software Engineer/client

 ✓ tests/lab-04/ActionsTaken.test.tsx (9) 869ms
 ✓ tests/lab-04/RequesterDashboard.test.tsx (1)
 ✓ tests/lab-04/StaffDashboard.test.tsx (2) 370ms
 ✓ tests/lab-04/TicketWorkflow.test.tsx (1)

 Test Files  4 passed (4)
      Tests  13 passed (13)
   Start at  09:06:15
   Duration  2.50s (transform 625ms, setup 384ms, collect 2.10s, tests 1.70s, environment 2.24s, prepare 907ms)
```